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Experience

Co-founder & CTO

Oct 2023 – Present

AlchemystAI (getalchemystai.com) – Bangalore, India

Raised \$700K+ (Antler, 100Unicorns, IPV)

Context Layer (solo-built)

- Designed and built the entire Context Layer from scratch – a deterministic context backbone for AI applications that gives any AI system durable memory and retrieval without vector database sprawl.
- Engineered graph-based fusion of vector similarity, keyword search, and graph traversal into a single retrieval pipeline, achieving deterministic results where RAG-only approaches produce inconsistent outputs.
- Implemented variance-based semantic continuity with a phase transition at **15–20 documents** – the system shifts from fixed-threshold to empirical variance, adapting retrieval behavior to corpus density automatically.
- Introduced consensus-free runtime scoping via composable group-name intersection, enabling hierarchical data isolation without schema migration – the key differentiator against Palantir Ontology at **~1/100th** the cost.
- Context Layer adopted by **9,500+** developers across **11+** Fortune 500 pilots and **2** enterprise deployments. Ranked **#1** upvoted DevTool on PeerList.
- Authored all developer documentation, API references, and integration guides from scratch at docs.getalchemystai.com.

Products & Revenue

- Architected and shipped **3 product lines** end-to-end: the Context Layer API (usage-based, billed by tokens), a Voice AI platform for B2B outbound calling automation (primary revenue driver), and an agent implementation services arm.
- Grew to **~\$400K ARR** in revenue through a pure B2B motion with zero brand trust – cold pipeline, enterprise pilots, and paid POCs only.

Models & Research (solo-built)

- Built and **open-sourced Pratilekha** – a family of Indic speech-to-text models for on-device ASR, released on HuggingFace under the AlchemystAI org. Name from Sanskrit: *prati* (in response to) + *lekha* (writing).
- Pratilekha ships in **3 variants**: v1 Tiny (**78.4MB**), v1.1 Tiny (**78.9MB**), and Small (**469MB**), covering **8** Indian languages – Bengali, Hindi, Kannada, Malayalam, Marathi, Odia, Tamil, and Telugu.
- Trained on KathBath, Common Voice, FLEURS, and IndicTTS datasets. Models validated at INT8 quantization; INT4 not recommended for Tiny variants due to quality degradation at that parameter scale.
- Built **Alchemyst C1** – a compact content-generation SLM targeting edge environments for summarization, entity extraction, and context synthesis within the Context Layer pipeline.
- Built **Alchemyst X1 (13.8GB)** – a large action model based on UC Berkeley's Gorilla v2 framework for tool use, function calling, and instruction following in autonomous agent workflows.

SDKs & Developer Ecosystem

- Published multi-language SDKs in TypeScript, Python, Go, Java, and Ruby on npm and PyPI.
- Built first-party integrations for LangChain, LlamaIndex, LiveKit, and OpenClaw.
- Managed SDK architecture migration from Stainless to Hey API using an additive subpath export strategy (@alchemystai/sdk/v1) with automated cross-repo GitHub Actions dispatch workflows for SDK regeneration on API spec changes.

Founder-led Sales, GTM & Media

- Led founder-led sales from day one – took discovery calls, scoped requirements, closed pilots, and converted paid POCs into recurring enterprise customers end-to-end before a dedicated sales team existed.
- Met directly with VPs and CXOs of Fortune 500 clients during early founder-led sales cycles, presenting technical architecture and business value in the same conversation.
- Transitioned from solo selling to operating alongside the sales and GTM team (CEO + COO), continuing to handle technical scoping, solution architecture, and fulfillment while co-founders own pipeline and relationships.
- Completed **500+** investor and customer pitches across the company's lifecycle – from pre-seed fundraising through enterprise pipeline development.
- Appeared as a guest on podcasts and media interviews discussing AI infrastructure, context systems, and the founder journey. Featured in Indian Startup Times, The Economic Times, and Financial Express.

Team & Operations

- Scaled engineering team from **0 to 10+**; established engineering culture, structured onboarding playbooks, and internal documentation at internal.getalchemystai.com.
- Built a voice client onboarding call playbook using a client typology framework (IDK / How / By When) with fallback phrases and escape hatches for junior runners.
- Managed internal burn while keeping unit economics stable – turned the voice business into a profitable line in a market where most AI startups are in a death spiral of losses. Context Layer operates at **80%+** gross margins.
- Navigated multiple exits and rough early-stage personnel challenges without team disruption or morale impact.

Infrastructure

- Led Qdrant vector database migration from GCP to AWS ap-south-1 (Mumbai) for Indian data residency compliance, upgrading v1.15 to v1.17.1 with resharding and timeout tuning – **3.75x** improvement in effective throughput.
- Integrated Trivy security scanning into Husky pre-commit hooks with graceful degradation for developer experience.
- Standardized ESLint v9 flat config and Turborepo monorepo structure across the voice platform codebase.

Full Stack AI Engineer

Memorang

2024
Remote, US

- Shipped production mobile applications built in **Expo + React Native** that generated **\$500K+** in revenue for the company.
- Worked across the full stack – React Native frontend, backend services, and ML model integration – delivering end-to-end features in a fast-moving product environment.

Core Developer, Applied AI

Arakoo.ai / EdgeChains (absorbed into Spring AI)

May 2023 – Oct 2023
Remote

- Built EdgeChains – a production-ready, JVM-based LLM SDK for enterprise, designed to run on existing Spring Boot infrastructure without a Python dependency.
- Developed scalable LLM caching technology to reduce redundant API calls and cut inference costs.
- Researched fine-tuning methods and prompt optimization strategies to reduce LLM usage costs at scale.
- Identified **17+** bugs across multiple projects in the Arakoo ecosystem; developed contributor documentation and onboarded new developers to the codebase.

Platform Maintainer

AI4Bharat / IIT Madras – Shoonya Platform

2022 – 2023
Remote, India

- Maintained Shoonya, the national-scale data annotation platform powering India's Bhashini project – the government initiative to build language technology across all **22** scheduled Indian languages.
- Contributed to open-source infrastructure for large-scale linguistic data collection, annotation workflows, and quality control pipelines.

ML Engineering Intern

CloudSEK – Digital Risk Protection

Sep 2022 – Dec 2022
Remote, India

- Worked in the Digital Risk Protection (DRP) department on the Fake Domain Finder product.
- Created and augmented a dataset for webpage elements, later used by the entire DRP team.
- Built a YOLOv7-based logo detection model to detect client logos from webpage screenshots, achieving **80%+** speedup over the pre-existing model with equal or better inference quality.
- Developed a gator model to filter out domain-sales pages, eliminating **20%+** false positives from the inference pipeline.
- Dockerized all models for testing and deployment by the DRP QA team.

Deep Learning Research Mentee

Google DeepMind

Jun 2022 – Jul 2022
Remote

- Selected as a mentee under a team of **3** researchers from Google DeepMind, North Carolina State University, and Carnegie Mellon University.
- Devised **10+** novel activation functions and their limitations in neural network use cases.
- Explored **2** novel activation functions that can overcome limitations of ReLU-family activations currently in widespread use.
- Ran **5+** benchmark tests and obtained performance metrics using scikit-learn, PyTorch, and Flux.jl.

Software Engineer (Contract)

Haplet LLC

Jun 2022 – Sep 2022
Remote, US

- Developed the FastAPI backend with **20+** endpoints for an MVP of a stealth product.
- Integrated the backend with a custom-trained YOLOv5 model to extract data from official documents using text and image detection.
- Created endpoints to mint the extracted data as NFTs by making API calls to blockchain services.
- Authored the design document for the entire backend architecture.

Publications

Roy, A., Roy, S.S. “ <i>Mitigating Hallucinations in LLMs Using Sieve of Fallacies and Truths (SoFT): A Game Theoretic Perspective.</i> ” Springer Nature Switzerland – In: Vajjhala et al. (eds), ISBN 978-3-031-76710-4.	2025
Roy, A., Sengupta, A., Pani, S. “ <i>Is a Large Context Window All You Need? Exploring TTFT–Context Size Tradeoff for Autoregressive LLMs.</i> ” engrXiv preprint (DOI: 10.31224/4666).	2025
Roy, A., Raj, S. “ <i>SerPyTor: A Distributed Context-Aware Computational Graph Execution Framework for Durable Execution.</i> ” arXiv preprint (abs/2304.07555).	2023
Roy, A. “ <i>Oscillating Activation Functions and Other Recent Progress in Activation Functions: A Survey.</i> ” engrXiv preprint (DOI: 10.31224/2407).	2022

Education

B.Tech, Computer Science & Engineering (IoT Specialization) Vellore Institute of Technology (VIT), Vellore	2020 – 2024 India
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Awards & Recognition

GDC Fellow 2022 – Selected top 7 out of 98,000+ applicants	2022
Impact Academy Graduate – AI Safety Mechanistic Interpretability Fellowship (Axiom)	2023
AWS Machine Learning Scholarship (Udacity)	2021
Gandalf Challenge (Lakera) – Solved both the Gandalf and Reverse Gandalf prompt injection challenges	
5,400+ GitHub Contributions with sharp acceleration in 2025	Ongoing
Press: Featured in Indian Startup Times, The Economic Times, Financial Express	2025–26

Technical Skills

Languages: Python, TypeScript/JavaScript, Go, Java, Rust, C/C++, Shell, R, SQL
AI/ML: PyTorch, scikit-learn, HuggingFace Transformers, Qdrant, LangChain, LlamaIndex, ONNX, TTS/STT pipelines
Infrastructure: AWS (EC2, S3, SQS), GCP, Docker, Kubernetes, Terraform, Turborepo, SIP/VoIP (Plivo, Exotel)
Backend: FastAPI, Django, Express, Spring Boot, PostgreSQL, MongoDB, Redis, Kafka, gRPC
Mobile: Expo, React Native
DevOps/Tooling: GitHub Actions, Trivy, ESLint v9, Husky, Stainless/Hey API (SDK generation), LiveKit

Open Source & Community

AlchemystAI GitHub Org (github.com/alchemyst-ai): SDKs in **5** languages, LangChain/LlamaIndex integrations, LiveKit & OpenClaw plugins, awesome-saas community project (**416** stars).
Contributor: Ivy/UnifyAI (ML framework unification), OWASP Maryam (OSINT), AI4Bharat/Shoonya (annotation platform).
Author: SerPyTor (distributed compute framework), Pymetrix (analytics library, **100+** GitHub stars), OpnEco (content suite, trending **#28** on ProductHunt).